Command-Line Administration

Why should I use CMD, when I can PWSH?

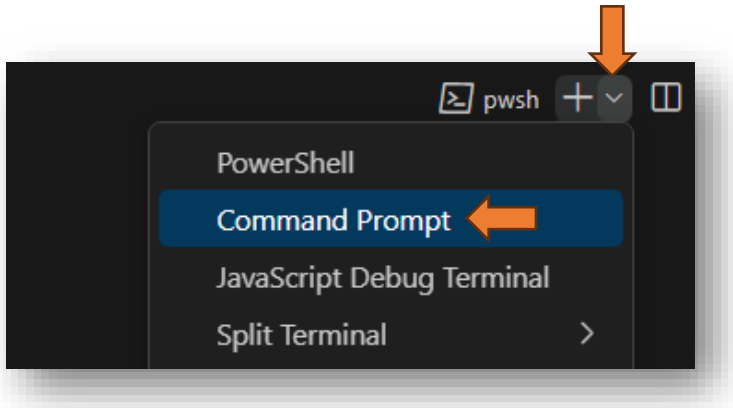
- Legacy, legacy, legacy…
  - Old scripts
  - Old systems
  - Old…people
- Some environments don’t support PWSH
  - WinPE…
  - …that is all…
- There are a few commands that have no PWSH replacement.
  - `GPRESULT`
- Some commands are just simpler, and faster that PWSH commands
  - `PING`
  - `NSLOOKUP`
- Understanding where we’ve come from will help you understand where we are now

Tips

- Working in CMD is *awful* (compared to PWSH), but you *should* get used to it.
- For this lab, here are a few things to make this easier:
  - ~~Use VSCode! CMD will not offer intellisense or highlighting of commands. VSCode and PWSH, however, will; BUT you should still get used to seeing CMD *in all it's glory!*~~
  - ~~Open VSCODE, create a .Bat file.~~
  - ~~Add a CMD prompt to the Terminal Window at the bottom.~~

OR

- use some sort of text editor (notepad++) to build your commands, then paste them into the console.



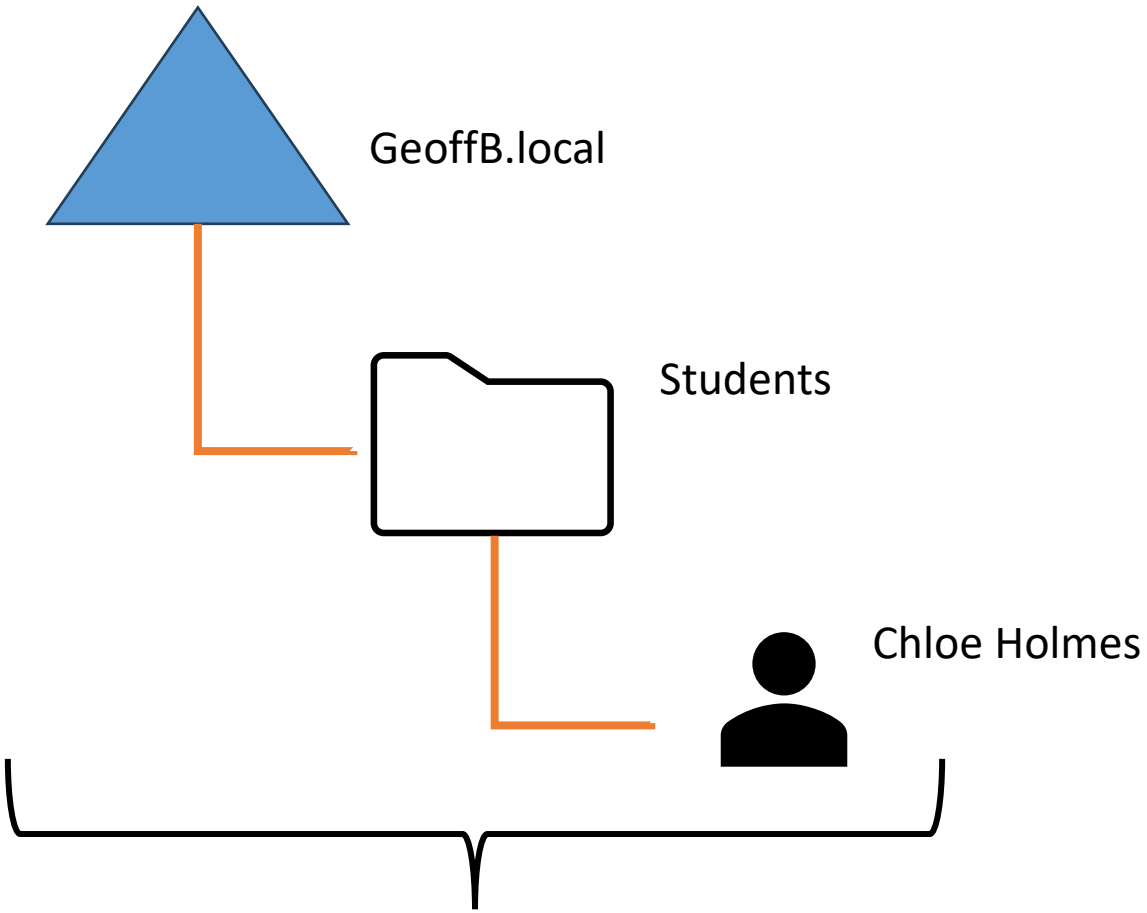
CMD AD MGMT

CMD Active Directory Management

LDAP Distinguished Names (review)

- LDAP names are broken into sections which identify the path within the database to the object
- LDAP names contain several keywords which help identify parts of an objects name:
  - **CN**
    - (Common Name)
    - Used to specify any leaf object (user, computer, group)
  - **OU**
    - used to specify any organizational unit objects
  - **DC**
    - (Domain Component)
    - any domains which the object is located within

Distinguished Name



CN=“Chloe Holmes”,OU=Students,DC=GeoffB,DC=Local

A note about history

- In PWSH I create profile-based functions so that I can quickly dump my history. This is useful for capturing that commands I used in a CMD session (for labs). Here's how it's done in CMD:
- To display history:
  ```
  DOSKEY /history
  ```
- To dump your history to a text file:
  ```
  DOSKEY /history > C:\historyfile.txt
  ```

- Note: your history is purged when you close your session window.

DSQUERY

(directory service query)

DSQUERY
(directory service query)

- A command that queries AD DS for list objects.

Syntax:

```
DSQUERY <object> <parameters>
```

- dsquery computer - finds computers in the directory.
- dsquery contact - finds contacts in the directory.
- dsquery subnet - finds subnets in the directory.
- dsquery group - finds groups in the directory.
- dsquery ou - finds organizational units in the directory.
- dsquery site - finds sites in the directory.
- dsquery server - finds AD DCs/LDS instances in the directory.
- dsquery user - finds users in the directory.
- dsquery quota - finds quota specifications in the directory.
- dsquery partition - finds partitions in the directory.
- dsquery * - finds any object in the directory by using a generic LDAP query.

DSQUERY

(directory service query)

- Examples:
  - `DSQUERY COMPUTER -name "Server*"`
  - `DSQUERY USER –samid “smattis”`
  - `DSQUERY USER –name “Sean Mattis” –o UPN`
  - `DSQUERY USER –name “Sean Mattis” –o DN`
  - `DSQUERY USER –name “Sean Mattis” –o RDN`
    (`-o` specifies output format)
  - `DSQUERY GROUP –name "DLG*"`

DSQUERY

(directory service query)

- Every DSQUERY object has other parameters, some of which are unique to that object type.
- Use DSQUERY /? for general
- Use DSQUERY <object> /? For help with that specific object.

Example: You can search for users by

```
DSQUERY user <DN>
DSQUERY user -name <username>
DSQUERY user -upn smattis@GeoffB.local
DSQUERY user -namep "Sean Mattis"
DSQUERY user -desc "The guy who needs his own
password policy"
```

DSADD

(directory service add)

DSADD.EXE

- A command-line program that enables you to create new AD objects, complete with attribute values
- Commonly used to create user accounts, groups, computer accounts and OUs
- When you have a large number of objects to create, the advantage of using dsadd.exe is that you can create a batch file (a script) containing multiple commands and create as many objects as you need at one time.
- The syntactical help works the same as DSQUERY

DSADD.EXE

- `dsadd computer` - adds a computer to the directory.
- `dsadd contact` - adds a contact to the directory.
- `dsadd group` - adds a group to the directory.
- `dsadd ou` - adds an organizational unit to the directory.
- `dsadd user` - adds a user to the directory.
- `dsadd quota` - adds a quota specification to a directory partition

DSADD OU

Create Organizational Units

DSADD OU

- To create an OU, the syntax is

```
dsadd OU <OUdn> -description “<desc>”
```

Do this…

- Create an OU called **MOD7** in the root of the domain
  `dsadd ou “ou=MOD7,dc=geoffb,dc=local”`
- Create and OU called MOD7**Students**
  `dsadd ou “ou=students,ou=`MOD7`,dc=geoffb,dc=local”`

DSADD USER

Create User accounts

DSADD USER

- DSADD USER has *a lot* of switches. Take your time with these commands.
- Perhaps write them in notepad first, then paste them into CMD. They can get very ugly and hard to read in the CMD window.

DSADD Switch	Meaning
-samid	Pre-Windows 2000 username
-fn	First Name
-ln	Last Name
-empid	Employee ID
-desc	Description
-office	Office
-email	Email address
-page	Pager Number
-fax	Fax Number
-webpg	Web Page
-dept	Department
-mgr <ManagerDN>	Manager
-profile	Profile Path
-mustchpwd {yes|no}	User Must Change Password
-reversiblepwd {yes|no}	Password stored using reversible encryption
-acctexpires{#Days}	Number of days until account expires

DSADD Switch	Meaning
-upn	User Principal Name
-mi	Middle Initial
-display	Display Name
-pwd {password | *}	Password; * will prompt for password
-memberof <groupDN>	Groups user belongs to
-tel	Telephone #
-hometel	Home Telephone #
-mobile	Cell Phone #
-iptel	IP Phone #
-title	Title
-company	Company
-hmdir	Home Directory
-loscr	Logon Script
-canchpwd {yes|no}	User Can change password
-pwdneverexpires {yes|no}	Password Never Expires
-disabled {yes|no}	Account is disabled

DSADD USER

- Syntax:
  dsadd user <userDN> [parameters]

Example:

```
DSADD USER “CN=Bird Person, OU=students,OU=MOD7,DC=GeoffB,DC=local”
–samid “bperson” –upn “bperson@GeoffB.local” –fn “Bird” –ln
“Person” –display “Bird Person” –desc “A bird like person” –pwd
‘Pa$$wOrd’ –hmdrv H: –hmdir \\Server1-GGB\Docshare\bperson -
disabled yes
```

```
dsadd succeeded:CN=Bird Person,OU=students,OU=PREP3.1,DC=GBEE,DC=local
```

Do this…

- Create a user called **Student1** in the **MOD7\Students** OU in the <your domain>.local domain
- Create a user called **Student2** in the **MOD7\Students OU**, with:
  - SAMID of **gmorris**
  - First name of **George**
  - Last Name of **Morris**
  - UPN of **George.morris@<your domain>.local**

DSADD GROUP

Create Domain groups

DSADD GROUP

- A command-line program that enables you to create new AD groups
- Syntax for creating group objects:
- `dsadd group <groupDN> [parameters]`
- **By default**, DSADD creates **global** security groups, but you can use command-line parameters to create groups with other types and scopes, as well as to specify members.

DSADD GROUP <parameters>

Parameter	Meaning
-secgroup {yes|no}	Specifies whether the program should create a security group (yes) or a distribution group (no). The default is yes.
-scope {l|g|u}	Specifies whether the program should create a domain local (l), global (g) or universal (u) group. The default is g.
-samid	Specifies the pre-Windows 2000 name of the group
-desc	Specifies a description for the group
-memberof <GroupDN>	Specifies the distinguished names of one or more groups which the new group should be made a member of
-members <objectDN>	Specifies the distinguished names of one or more objects that should be made members of the new group.

Do this…

- Create a global group:
  - called MOD7Students
  - in the MOD7 OU
  - with a description of “Group for Students”
  - and has student1 for a member.

DSADD Computer

Create computer accounts

DSADD Computer

- To create a computer account, the syntax is:
- `dsadd computer <computerDN> [parameters]`
- DSADD Computer Parameters:

DSADD parameter	Meaning
`-samid`	Pre-Windows 2000 account name
`-desc`	Description
`-loc`	Location
`-memberof <GroupDN>`	Group(s) computer should be a member of

DSADD COMPUTER example

- Dsadd computer “CN=WebServer1,CN=Computers,DC=GeoffB,DC=LOCAL” –desc “This computer doesn’t exist!”

DSMOD

A command-line tool that you can use to modify existing AD objects.

DSMOD USER

- The syntax and command-line switches for modifying user objects with `Dsmod.exe` is *exactly the same* as with `Dsadd.exe`
- Except: you cannot use the `-samid` or `-memberof` switches

- Syntax:
- `dsmod user <userDN> [parameters]`

DSMOD GROUP

- A command-line tool that you can use to modify existing AD groups.
- Syntax for modifying existing groups:

```
dsmod group <groupDN> [parameters]
```

DSMOD GROUP Parameters

Parameter	Meaning
`-secgroup {yes|no}`	Specifies whether the program should create a security group (yes) or a distribution group (no). The default is yes.
`-scope {l|g|u}`	Specifies whether the program should create a domain local (l), global (g) or universal (u) group. The default is g.
`-addmbr <members>`	Adds members to the group. Replace <members> with the DNs of one or more objects.
`-rmmbr <members>`	Removes members from the group. Replace <members> with the DNs of one or more objects.
`-chmbr <members>`	Replaces the complete list of group membership with the DN's specified as <members>

Do this

- Modify Student 2 so their first name is ‘Sam’
- Add Student 2 to the PREPStudents Group.

DSMOD COMPUTER

- Syntax:
  `dsmod computer ComputerDN [parameters]`
- Additional DSMOD Computer Parameters

-disabled {yes|no}	Disables or enables computer account
-reset	Resets the computer account
-desc	Changes the description
-loc	Changes the location

DSMOVE

Moves existing Active Directory Objects

DSMOVE

- Rename or Move an active directory object (user, computer, group) to a different Organizational Unit (OU).
- Syntax:
  `DSMOVE ObjectDN [newparent ParentDN] [newname NewRDN]`

  `-newRDN` is for renaming the object during the move.

- **Note: "RDN" is** the *Relative Distinguished Name (RDN*) is simply the first **component of the distinguished name**.

Do this…

- Add a new user named ‘teacher’ in the ‘Users’ container.
- Move the ‘teacher’ user from the ‘Users’ container to the MOD7 OU

DSRM

Removing objects from Active Directory

DSRM

- Command-line utility used to delete objects from Active Directory
- Syntax:
- `DSRM ObjectDN [-subtree [-exclude]] [-noprompt] [{-s Server | -d Domain}] [-u UserName] [-p {Password | *}] [-q] [{-uc | -uco | -uci}]`
  - `-subtree` Delete the object and all objects contained in its subtree.
  - `-exclude` Delete all objects contained in the subtree, but not the object itself.
  - `-noprompt` Do not prompt to confirm deletion.
  - `-s` Server Connect to a remote server/domain, default=%logonserver% domain controller.
  - `-c` Continue with the next object after any error (when you specify multiple target objects) **by default dsrm will *exit* when the first error occurs.**
  - `-q` Quiet, suppress all output

Do this…

- Delete the ‘teacher’ user

NET.EXE

A command-line tool that you can use to modify many aspects of a Windows Server

Has several modes that can be accessed by following the NET command with the mode to enter

NET SHARE

- NET SHARE makes a server's resources available to network users.
- When used without options, it lists information about all resources being shared on the computer.
- For each resource, Windows reports the device name(s) or pathname(s) and a descriptive comment associated with it.
- To share a folder, use the following syntax:

```
NET SHARE <sharename>=drive:path [/USERS:number |
/UNLIMITED] [/REMARK:"text"] [/GRANT:user,[READ | CHANGE
| FULL]]
```

- To stop sharing a shared folder, use the following syntax:

```
NET SHARE {sharename | devicename | drive:path} /DELETE
```

NET USE

- `NET USE` is a command line tool used to map network drives to shared network folders
- To map a drive using the `NET USE` command, use the following syntax:

```
NET USE [driveletter]:[UNCPath]
[/USER:[domainname\]username] [/PERSISTENT:{YES | NO}]
```

- To delete a mapped network drive, use the `NET USE` command with the following syntax:

```
NET USE [driveletter | UNCPath] /delete
```

NET VIEW

- `NET VIEW` displays a list of resources being shared on a computer. When used without options, it displays a list of computers in the current domain or workgroup.
- To view a system or domain's shared resources, use the syntax:

```
NET VIEW [\\computername | /DOMAIN[:domainname]]
```

Do this…

- Using CMD and net commands display all Shares on the local PC. Create a folder called “**C:\DeleteMe**”.
- Share that folder with a share name of “BadShare” using `net share`.
- Map the **C:\DeleteMe** folder to **F:** using `net use`.
- Unmount the share using `net use`.
- Delete the share using `net share`.

Mr. Beer’s non-comprehensive list of useful command line commands that have no Powershell equivalent or are just too convenient to stop using…

Yeah!

Title!

- `ipconfig [/all]` – show logical addressing of attached interfaces.
- `ipconfig [/release | /renew]` – Release a DHCP lease and obtain a new one.
- `ipconfig [/flushdns] [/displaydns]` – Clear/View the local DNS cache
- `ping [-t]` – ICMP test for connectivity. `[-t]` for an infinite ping (to confirm remote reboots)
- `gpupdate [/force]` – apply all GPO settings for user and computer
- `arp [-a]` – display the local ARP cache. Find machines by MAC/OUI, etc.
- `shutdown [/r] [/t 0]` – Restart the computer. `[/r]` for restart, `[/t 0]` to wait zero seconds.
- `Netstat [-a] [-n] [-o]` – display open connections and listening ports. `[-a]` is all ports (including inactive), `[-n]` display IP address instead of DNS names, `[-o]` ID of the process that owns the connection.
- `Gpresult /h report.html /f` – print out a report of the applied GPOs for a computer and the current user.
- `net user <username> /domain` – show a user's account info in AD.

Honorable Mentions

- `dcdaig` – analyses the state of DCs in a forest. Troubleshooting tool with various diagnostic tests.
- `netdom` – for managing AD domains and trusts between domains. Largely replaced by PWSH.
- `ntdsutil` – Perform database maintenance of AD. Outside the scope of this course.